BYTECUBIT TECHNOLOGIES · INDUSTRIAL AUTOMATION ENGINEER

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Industrial Systems Modernization | Manufacturing Data & Automation

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90-Day Entry Plan

Stabilize the estate, prove one bounded migration, and leave a reusable release method.

ENTRY PRINCIPLES

Production first

Do not treat a live plant as a development environment. Bound risk, observe behavior, and preserve rollback.

Evidence before scale

One well-proven reusable pattern is more valuable than several fast bespoke screens.

Context is part of data

Names, units, timestamps, quality, asset relationships, ownership, and intended use travel with the signal.

Adoption is engineering

Operator workflow, diagnostics, support, training, and decision rights are part of application design.

DAY-ONE QUESTIONS

- Which production process and function creates the most costly recurring friction?
- What is the Ignition topology, module mix, environment structure, and release path?
- How are tags, UDTs, scripts, queries, projects, and credentials governed?
- Which MES/PI/SQL dependencies are documented, and which are only known locally?
- Who has authority to accept, hold, cut over, and roll back?

INITIAL OUTPUTS

Application/historian/data estate map · Stakeholder and decision-rights map · Risk and dependency register · Candidate pilot selection · Baseline support and data-quality evidence · Development/release convention review.

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Days 1-60

Read the system, stabilize immediate risks, then build the first target twin.

DAYS 1-15 · ORIENT AND TRIAGE

Work	Output	Evidence
Walk the process with operators, controls, application, quality, maintenance, R&D, and data users.	Workflow and stakeholder map	Observed states, handoffs, workarounds, and consequences
Inventory Ignition, legacy MES, SQL, PI, interfaces, environments, and support ownership.	System estate and dependency map	Projects, tags, queries, points, interfaces, consumers, owners
Review recurring issues and urgent risks.	Stabilization backlog	Issue frequency, downtime/rework categories, diagnosis gaps

DAYS 16-30 · DEFINE THE METHOD

- Select one bounded pilot with real value, manageable consequence, and representative architecture.
- Establish source control, naming, UDT/tag, component, SQL access, testing, release, and rollback conventions.
- Define the Signal Twin acceptance record and observation window.
- Resolve immediate PI/SQL data-quality issues that block the pilot.

DAYS 31-60 · BUILD AND RECONCILE

- Develop the target Ignition component and diagnostics.
- Map queries, transactions, historian points, contexts, and downstream consumers.
- Run target behavior against representative events and exceptions.
- Conduct operator, manufacturing, quality, support, and R&D reviews.
- Prepare cutover, rollback, and post-release observation.

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Days 61-90

Release the proven function and turn the pilot into a repeatable modernization system.

DAYS 61-75 · RELEASE DELIBERATELY

- Complete parity, exception, performance, usability, access, and rollback evidence.
- Conduct release-readiness review with named decision owner.
- Cut over during an agreed production window.
- Observe events, data quality, operator use, support signals, and downstream consumers.
- Roll back immediately if pre-agreed thresholds are breached.

DAYS 76-90 · COMPOUND THE PATTERN

- Publish reusable Ignition component, tag/UDT, query, diagnostic, test, and release patterns.
- Update PI naming/context/stewardship guidance based on observed work.
- Define the cloud data-product contract: schema, semantics, quality, freshness, access, owner, and consumer.
- Prioritize the next tranche by operational value, dependency risk, reuse potential, and readiness.
- Establish weekly modernization review and monthly architecture/data stewardship review.

90-DAY SCORECARD

Outcome	Evidence at day 90
Production value	One bounded function released or a documented hold decision supported by evidence.
Modernization method	Signal Twin acceptance record, release/rollback convention, reusable patterns.
Data trust	Documented tag/SQL/PI context and consumer map for the pilot domain.
Supportability	Diagnostics, runbook, owner, issue path, and post-release observation.
Portfolio direction	Prioritized next tranche with value, risk, dependencies, and standardization opportunities.

EXTENSION CASE

> The strongest extension argument is not effort expended. It is a safer, faster, repeatable modernization method with one proven release and a credible pipeline of next work.